Power BI

INTERMEDIATE

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SUMMARY


IMPORTING THE DATA
Power BI
Desktop


BASIC DAX MEASURES
Power BI
Desktop


Creating the pages
Power BI
Desktop


CROSS-HIGHLIGHTING / FILTERING
Power BI
Desktop


Combo chart
Combining lines & columns + average
Power BI
Desktop


BOOKMARKS
Power BI
Desktop

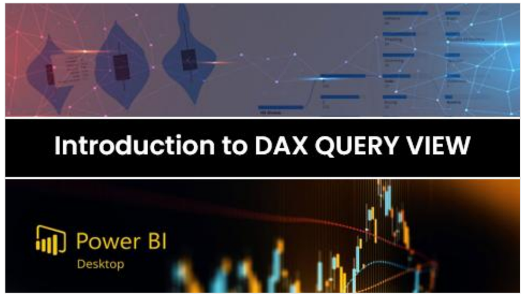

FIELD PARAMETER
Power BI
Desktop

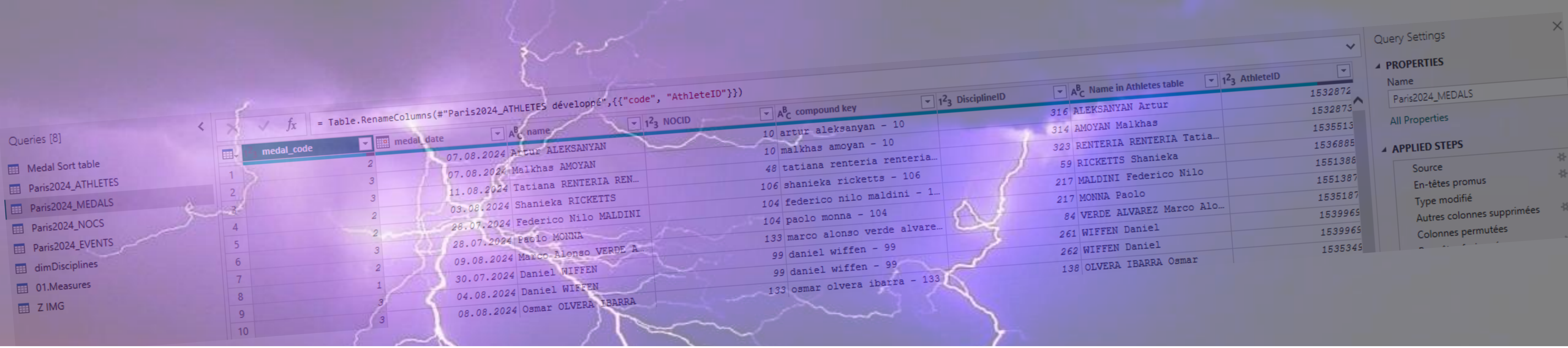

Visual DAX
Power BI
Desktop


Calculation groups
Power BI
Desktop


DRILLTHROUGH PAGE
Power BI
Desktop


TOOLTIP PAGE
Power BI
Desktop


Introduction to DAX QUERY VIEW
Power BI
Desktop



IMPORTING THE DATA

Connecting to data sources and cleanup

Import the nocs.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/nocs.csv

- Promote headers.
- Keep only these columns: [code],[country_long].
- Add an index column, rename it [NOCID] and place it in 1st position.
- Check the column types.

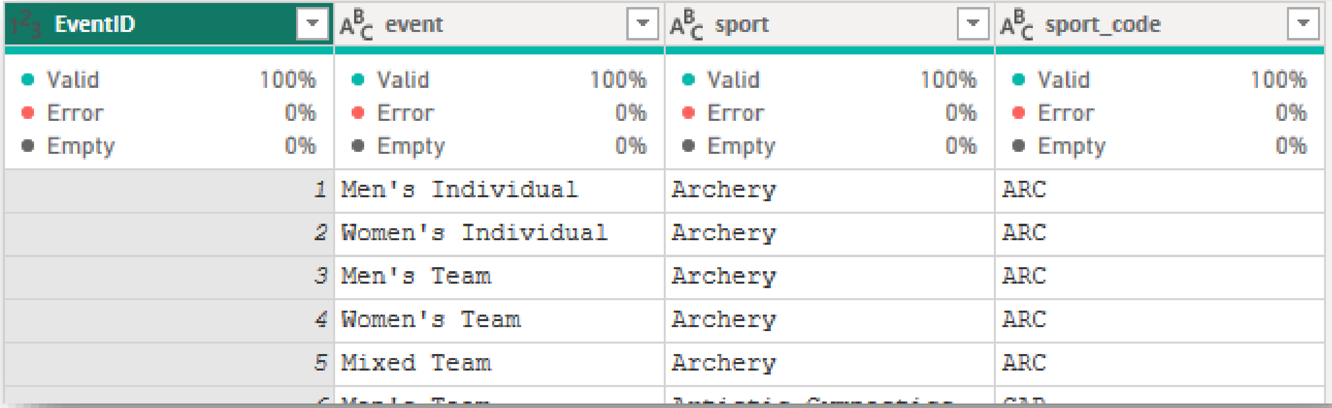
	NOCID	code	country_long
1	1	AFG	Afghanistan
2	2	AHO	Netherlands Antilles
3	3	AIN	AIN
4	4	ALB	Albania
5	5	ALG	Algeria
6	6	AND	Andorra
7	7	ANG	Angola
8	8	ANT	Antigua and Barbuda

Connecting to data sources and cleanup

Import the events.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/events.csv

- Promote headers.
- Keep only these columns: [event], [sport] and [sport_code].
- Add an index column, rename it [EventID] and place it in 1st position.
- Check the column types.



EventID	event	sport	sport_code
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	Men's Individual	Archery	ARC
2	Women's Individual	Archery	ARC
3	Men's Team	Archery	ARC
4	Women's Team	Archery	ARC
5	Mixed Team	Archery	ARC

Connecting to data sources and cleanup

Import the athletes.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/athletes.csv

- Promote headers.
- Keep only these columns: [code],[name0_tv],[gender],[country_code].
- Merge the table with the NOCS table to insert the [NOCID] field on each row.
- Remove [country_code].
- Check the columns types.

code	name_tv	gender	NOCID
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1532872	Artur ALEKSANYAN	Male	10
1532873	Malkhas AMOYAN	Male	10
1532874	Slavik GALSTYAN	Male	10
1532944	Arsen HARUTYUNYAN	Male	10
1532945	Vazgen TEVANYAN	Male	10

Connecting to data sources and cleanup

Import the medals.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/medals.csv

- Promote headers.
- Keep only these columns: [medal_code],[medal_date],[name],[discipline],[event],[country_code].
- Merge the table with the Athletes table to insert the [code] field on each row and delete [country_code]. Use [name] and [name_tv] for merging.
- Merge the resulting table with the Events table to add the [EventID] key column. Use both [discipline] and [event] columns as the key columns for merging.
- Check the columns types.

Connecting to data sources and cleanup

Import the medallists.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/medals.csv

- Promote headers and filter [is_medallist] to True.
- Keep only these columns: [medal_code],[medal_date],[gender],[discipline],[event],[birth_date] & [code_athlete].
- Merge the table with the Events table to insert the [EventID] field on each row and delete [discipline] and [event].
- Add a new column that calculates the age (in days) by substracting the birth date from today's date.
- Check the columns types.

Creating helping tables

1. In the Home tab, click on 'Enter data' then give the new empty table a name (01.Measures for example) and click 'OK'.
   This table will be used to store all the DAX measures needed for this report.
2. Create another manual table that will store the IMG URL that will be included in your pages. Name it 'Z.IMG' for example.
   Add 2 columns [ID] and [URL] and fill the first row with the official Paris 2024 Logo picture. Repeat for the LA2028 logo and whatever picture you widh to include in your report.
3. Insert a new table and call it 'Medal Sort Table'. Insert a column for the type of Medal, another one for the Sort order and a third one that contains the medal_code info matching the values in the Paris2024_MEDALS table.

Create the data model

Model View

1. In the model view, check the detected relationships and add or delete as needed.
2. Add a Calendar table by inserting the DAX script provided for this purpose.
   Name it 'CALENDAR'.
   Configure the start and end dates in the DAX code.
   Mark as a date table and select the date field in the calendar table as the date reference.
3. Create a 'hr_date' hierarchy in the CALENDAR table and add [Year], [Month], [Week Number], and [Date].
   Don't forget to sort [Week Number] by the [Week] column.

BASIC DAX MEASURES

DAX MEASURES

Create basic explicit DAX measures such as:

```
NB Medals = COUNTROWS(Paris2024_MEDALS)

NB GOLD = CALCULATE(COUNTROWS(Paris2024_MEDALS),
              Paris2024_MEDALS[medal_code]=1
)

NB SILVER = CALCULATE(COUNTROWS(Paris2024_MEDALS),
              Paris2024_MEDALS[medal_code]=2
)

NB BRONZE = CALCULATE(COUNTROWS(Paris2024_MEDALS),
              Paris2024_MEDALS[medal_code]=3
)

NB days = DISTINCTCOUNT(Paris2024_MEDALS[medal_date])
```

DAX MEASURES

Create basic explicit DAX measures calculating the overall average number of medals:

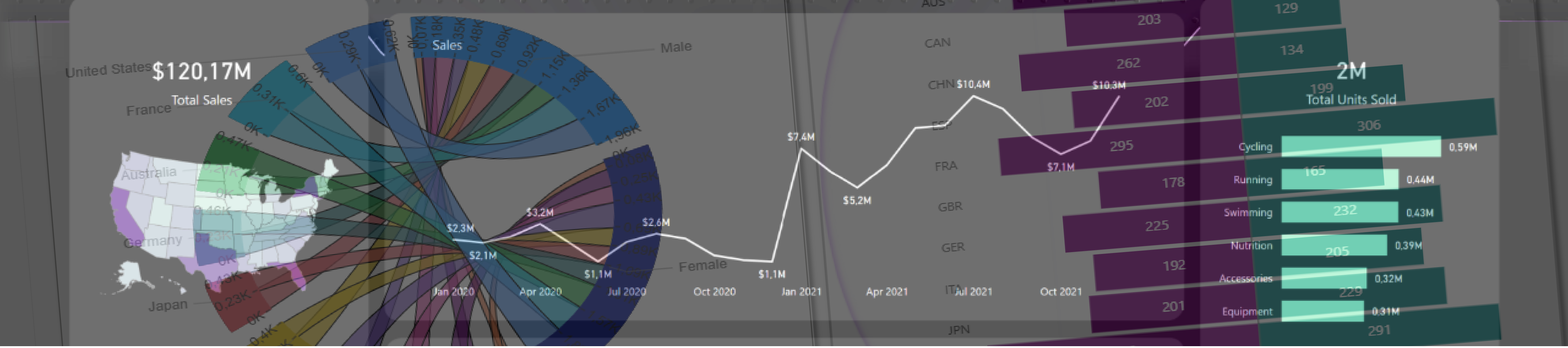
```
AVG medals per day (global ) =

VAR __totdays = CALCULATE([NB days],
                          REMOVEFILTERS(Paris2024_MEDALS))

VAR __totmedals = CALCULATE([NB Medals],
                          REMOVEFILTERS(Paris2024_MEDALS))

RETURN

ROUNDDOWN(DIVIDE(__totmedals,__totdays),0)
```



Creating the pages

Start Page

- PURPOSE: To provide a start page with a navigation menu to all pages in the report.
- Format the background of page 1 with a Scrim (BigSquare.png).
  Page Size>Canvas Background> Image
- Scrims available here:
  https://github.com/artofnet-training/PBI-INT/SCRIMS/SCRIMS-BLUE.zip
- Download Markus Kolp's 'Image' visual and insert it into the page. In "Image URL", drag Z IMG[URL] and apply a filter on this visual on the Z IMG[ID] field in order to select the Paris2024 logo to be displayed.
- Insert a page navigation menu. Insert > Button > Browser > Page Browser.
- Insert a card visual displaying the remaining days till LA2028 (create a DAX measure) and insert the LA2028 logo in format > Image > image URL.
  The Dax measure to compute the remaining days:
  Countdown LA2028 = INT(TODAY()-DATE(2028, 7, 14)) & " days"

The Athletes Page

1. Create three measures calculating the total number of athletes and the number of male/female athletes.
   NB athletes = COUNTROWS(Paris2024_ATHLETES)
   NB Men = CALCULATE(COUNTROWS(Paris2024_ATHLETES), Paris2024_ATHLETES[gender]="Male")
   NB Women = CALCULATE(COUNTROWS(Paris2024_ATHLETES),
   Paris2024_ATHLETES[gender]="Female")

2. Insert these 3 measures into cards at the top of the page.
3. Insert a matrix with Paris2024_NOCS[country_long] as "Lines" and [NB athletes] as "Values".
4. Add conditional formatting for the values.
   Format> Cell elements > Series: NB Athletes > Data Bars(turn on)
5. If you can, download a visual from the store. To do this, click the three dots 'Get more visuals' at the end of the list of visuals in the 'Build' panel.
   Once in the store, search for Microsoft's 'tornado chart' and download.
6. Insert a Tornado visual in the page with NOCS[code]in "Group", Paris2024_ATHLETES[gender] in "Legend" and [NB athletes] in "Values".
7. Add a split screen scrim for background

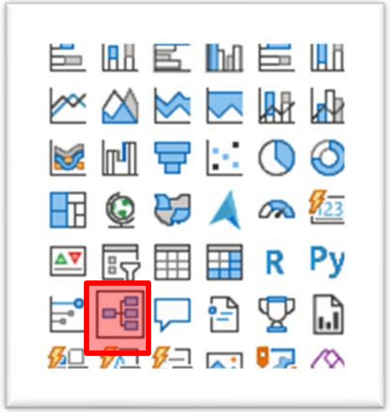
The ‘Athletes’ page

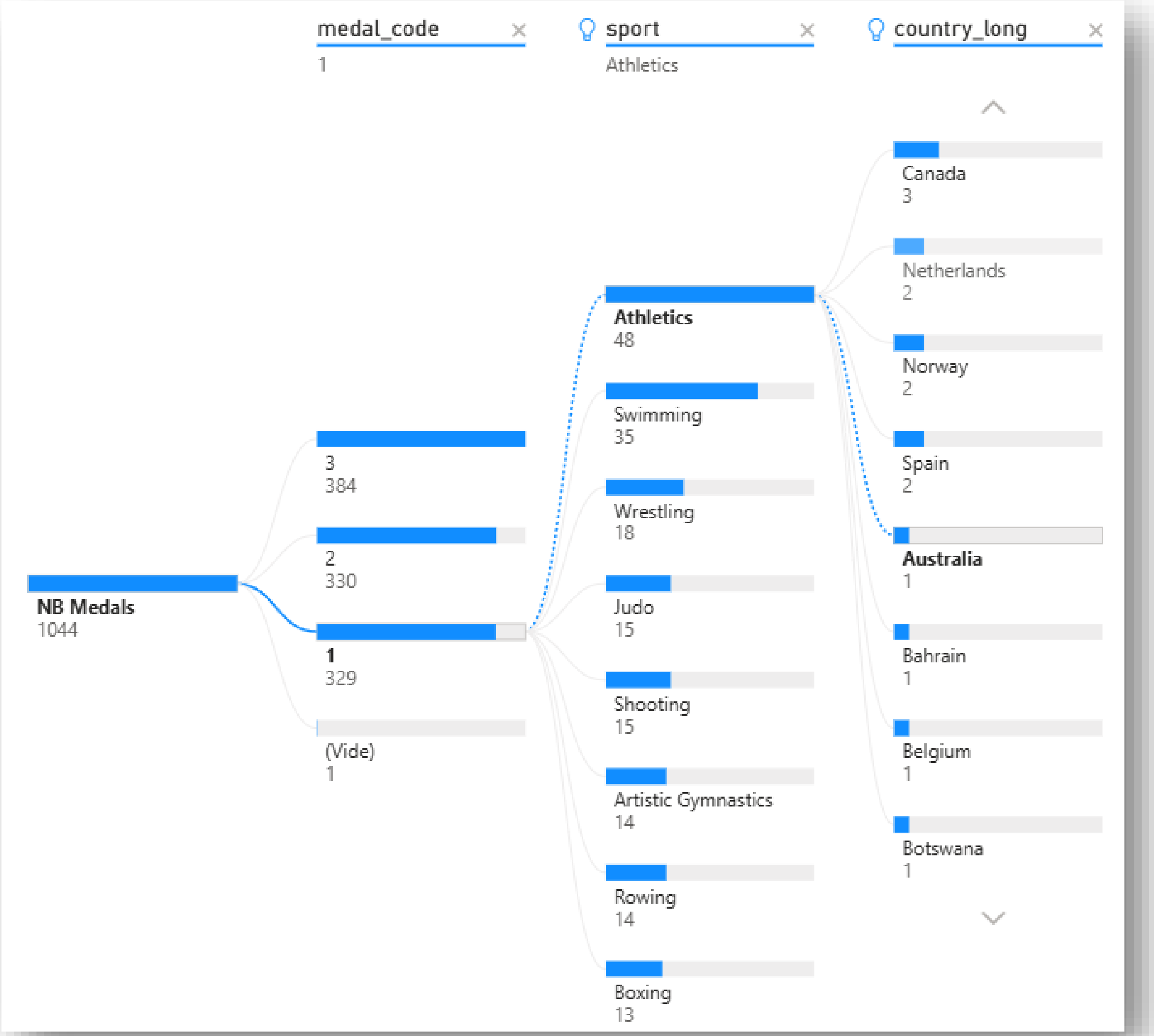
country_long	NB athletes
Argentina	143.0
Australia	474.0
Belgium	177.0
Brazil	290.0
Canada	332.0
Egypt	155.0
France	601.0
Germany	457.0
Great Britain	343.0
Hungary	177.0
Ireland	143.0
Italy	397.0
Japan	430.0
Netherlands	290.0
Total	**6406.0**

Create a decomposition tree

In a new page, insert a 'decomposition tree' visual and drag [NB Medals] into the 'Analyze' well.

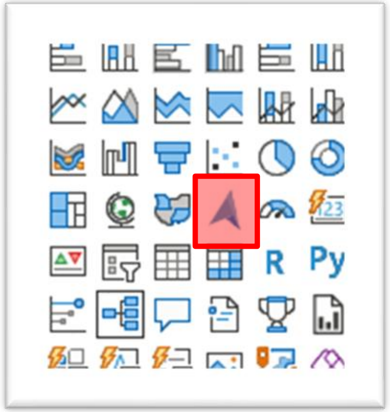
Drag Medal Sort Table[Medal], Paris2024_NOCS[country_long], Paris2024_EVENTS[sport], and Paris2024_EVENTS[event] into "Explain by."

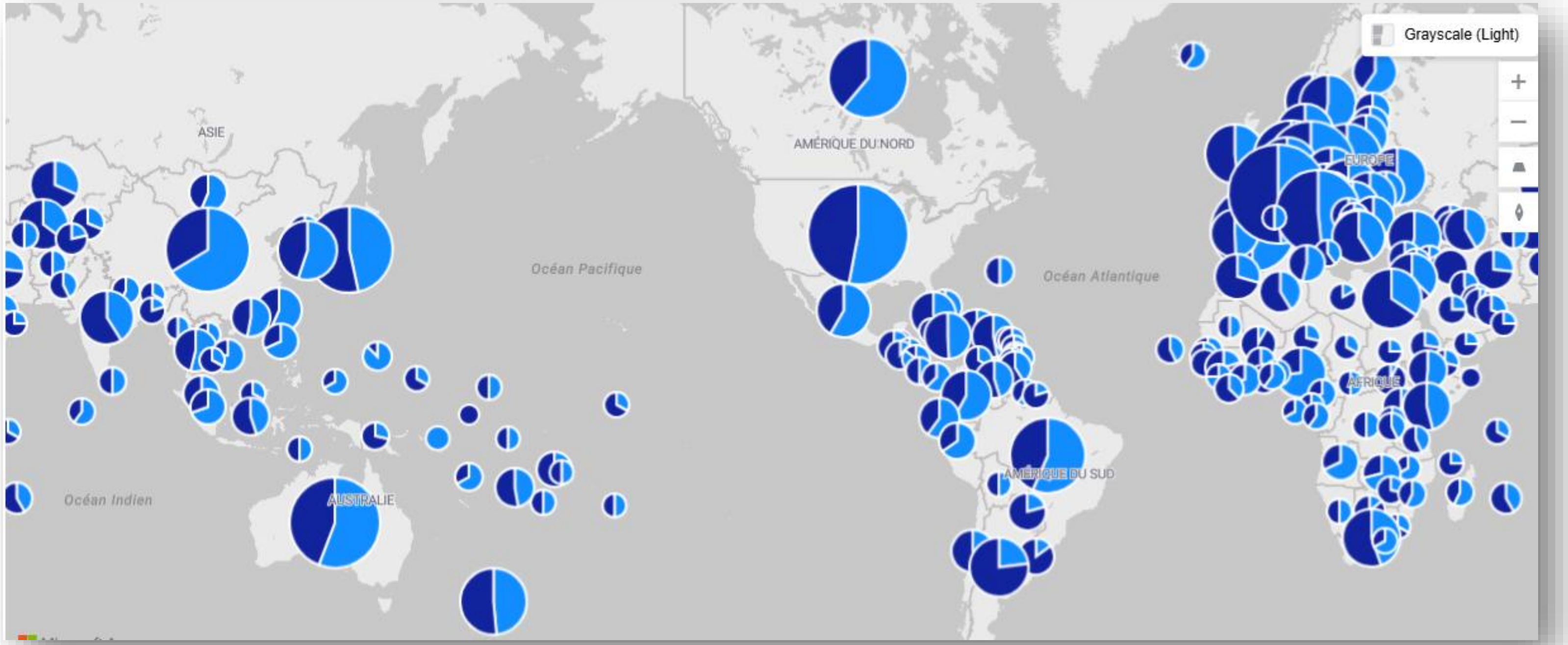




Create an Azure Maps visual

Insert an Azure Maps visual and drag Paris2024_NOCS[country_long] into "Location", [NB athletes] into "Size" and Paris2024_ATHLETES[gender] into "Legend".



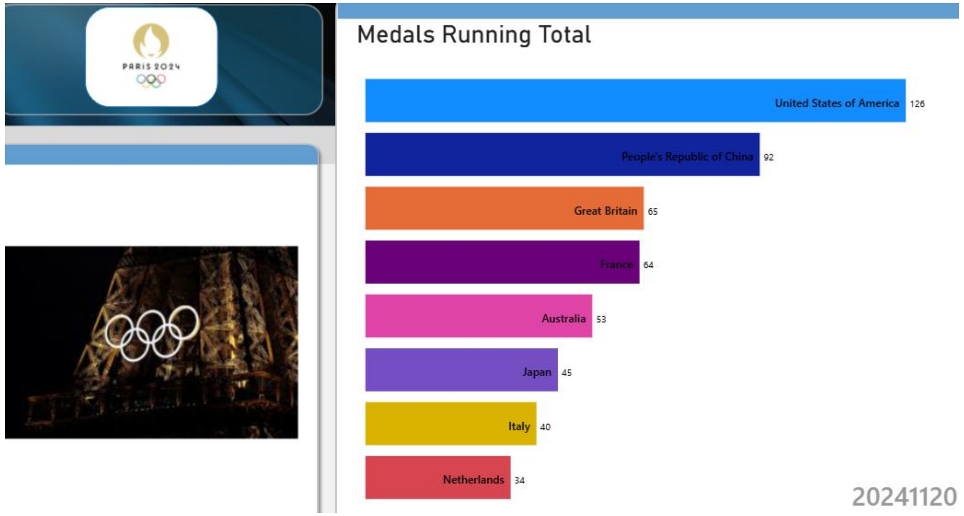


Animated bar chart

- Download a new 'animated bar chart race' (designed by Wishyoulization) from the store and insert it in a new page.
- Create a DAX measure that computes the Running sum of medals per day:

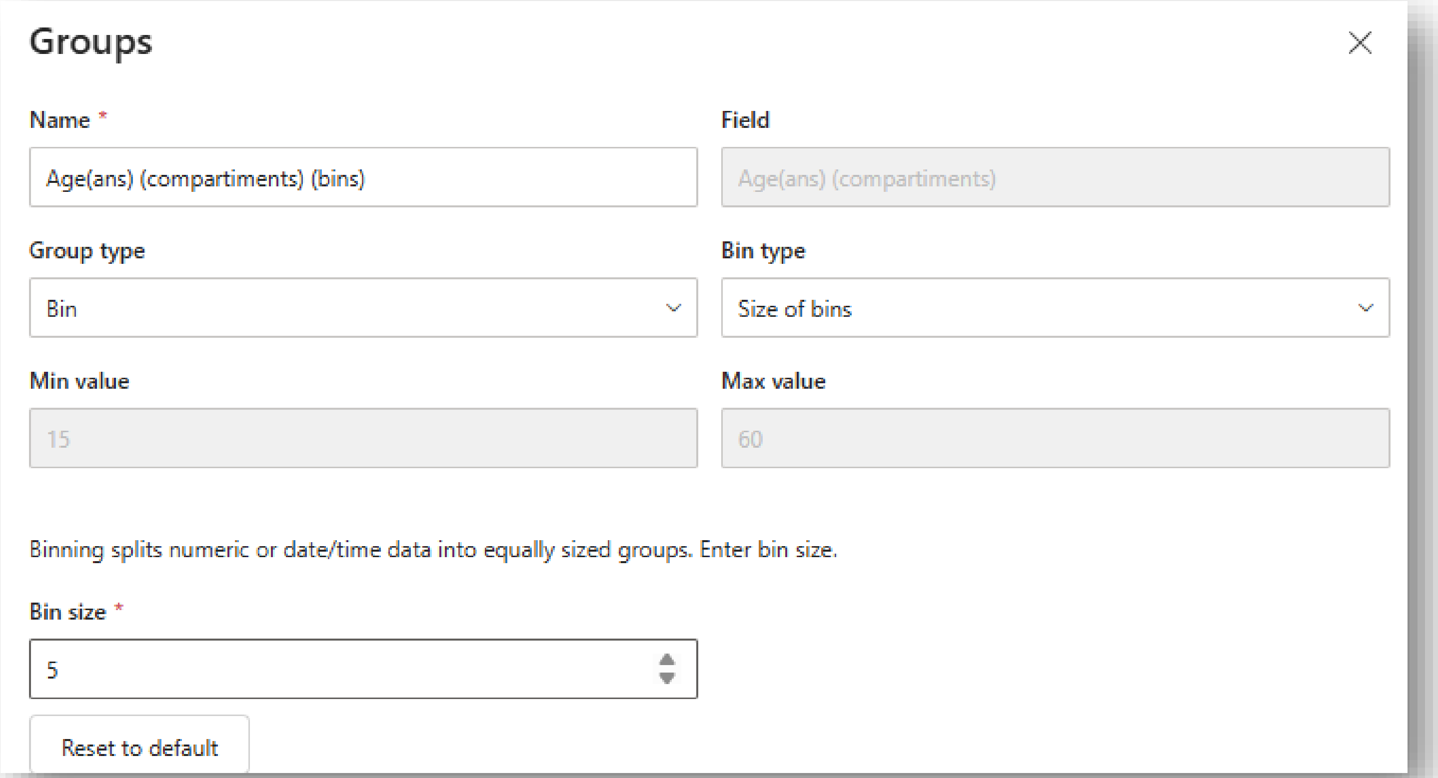
```
Running Total Medals =  CALCULATE(
                            [NB Medals],
                            'CALENDAR'[Date] <= MAX('CALENDAR'[Date]))
```

- Add Paris2024_NOCS[country_long] in "Name", [Running Total Medals] in "Value" and Calendar[Date as integer] in "Period" (period must be numeric for display).
- Go to Format > Visual Object > Title and insert the title of your choice.
- Go to Format > Visual Object > Wishyoulization Settings > Duration and insert 2000 or more, enable the 'hideGrid' option and enter 32 in 'periodSize'. If necessary, change the value of 'TopN' to display fewer results during the animation.
- Double click on the visual to start the animation.



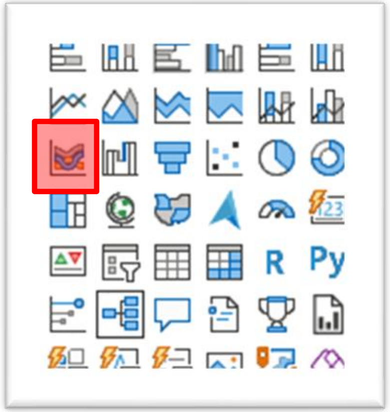
Ribbon chart with age ranges

- Create the following DAX metric:
  NB Medallists = COUNTROWS(Paris2024_MEDALLISTS)
- Add a calculated column in Paris_2024_MEDALLISTS:
  Age (y) = INT(DIVIDE(Paris_2024_MEDALLISTS[Age],365.25))
- Right-click on Age (y) in the 'Data' panel and select 'New Group'.
- Choose a bin size of 5



Ribbon chart

- Insert a ribbon chart with [age (y)] in "X-axis", [NB Medallists] in "Y-axis" and Medal Sort table[Medal] in "Legend". View Data Labels (Format)Visual Object > Data Labels and adjust the density in the same option group as desired. Change the title.
- Filter the ribbon chart on [age(y) (bins)] and keep values between 15 and 40.
- Insert a matrix visual with [age (y) (bins)] in "Rows", [NB Medallists] in "Values" and Medal Sort table [Medal] in "Columns".
- Insert a segment on Paris2024_ATHLETES[gender].
- Insert 3 cards that display [NB Medallists] as the grand total + number of F and M medallists.
- Disconnect the slicer from these 3 total cards.



The 'Medallists' page

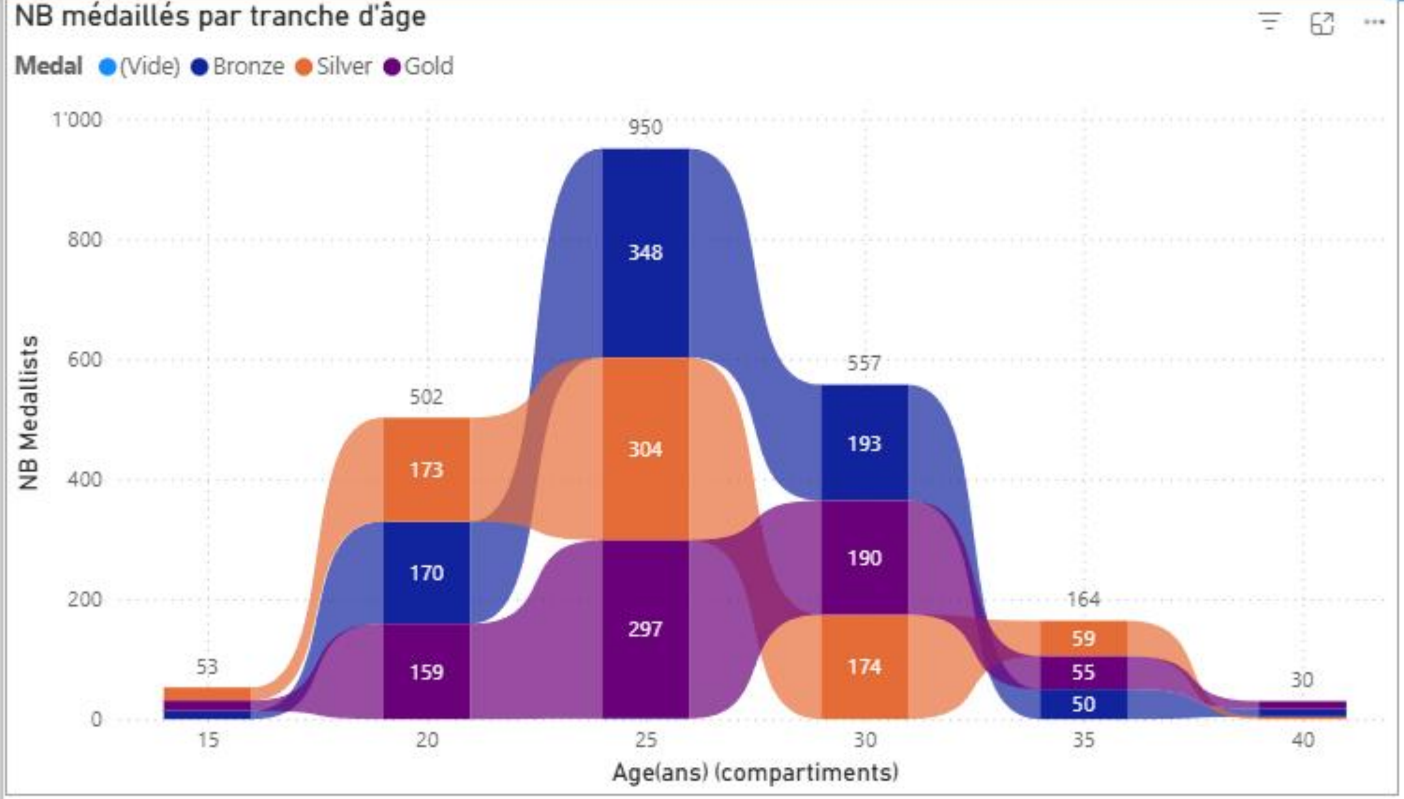
NB Medallists
2.268K

Médaillés F
NB Medallists
1.142K

Médaillés M
NB Medallists
1.126K

☐ Female
☐ Male

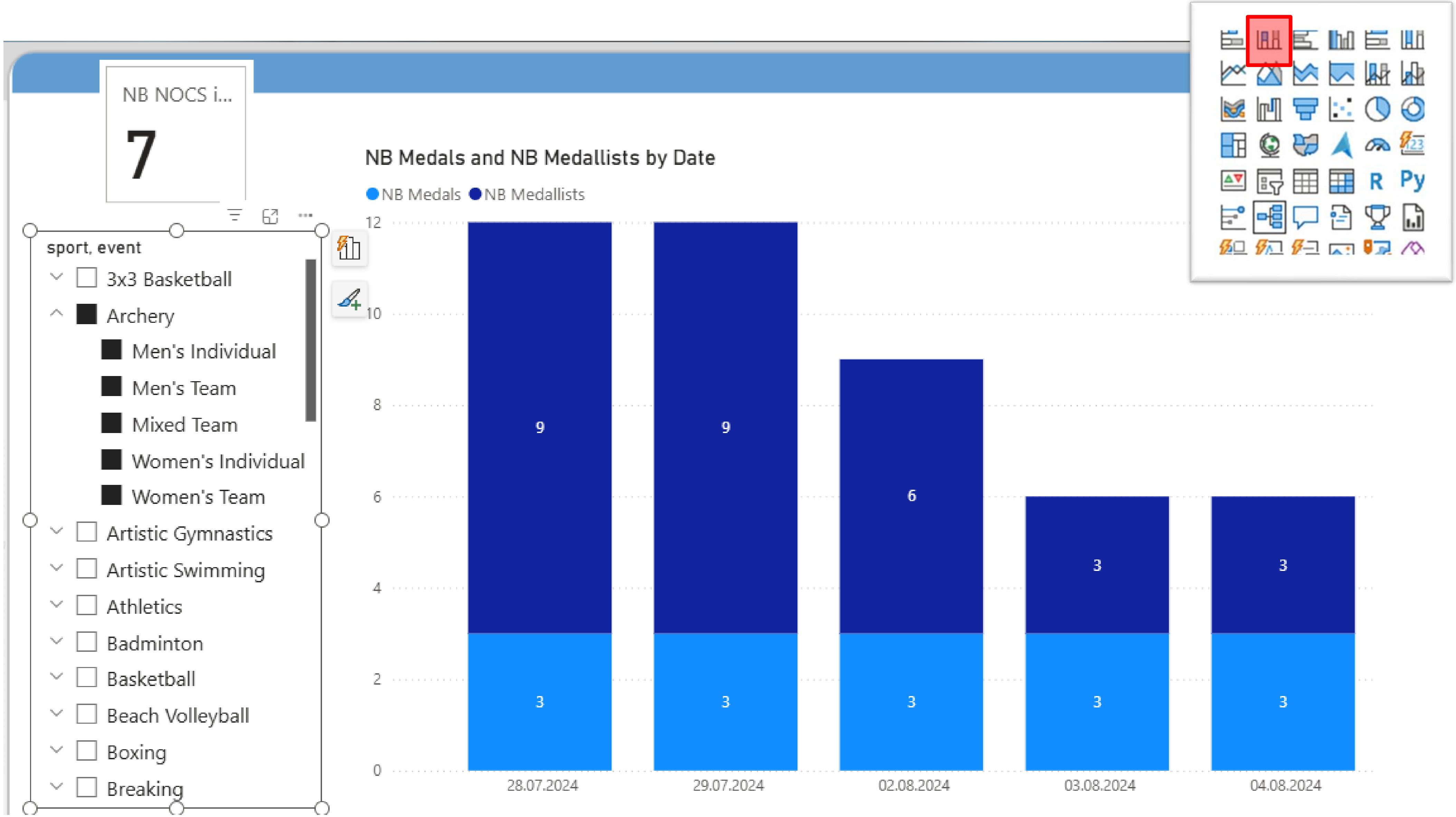
Age(ans) (compartiments)		Bronze	Silver	Gold	Total
15		14	22	17	**53**
20		170	173	159	**502**
25	1	348	304	297	**950**
30		193	174	190	**557**
35		50	59	55	**164**
40		13	4	13	**30**
45		3	1	1	**5**
50			3		**3**
55		1	1	1	**3**
60			1		**1**
Total	**1**	**792**	**742**	**733**	**2268**



Stacked Histogram

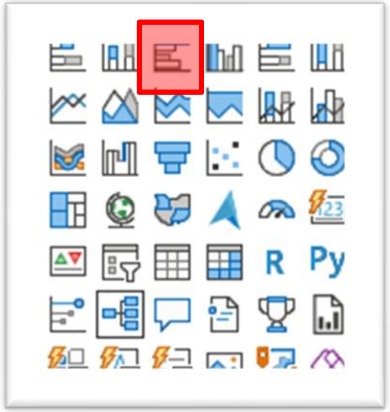
- Insert a stacked histogram with Calendar[Date] in "X-Axis" and [NB Medals] + [NB Medallists] in "Y-Axis".
- Enable data labels and adjust the density (Format > Data Labels > Options > Label Density).
- Insert a slicer and drag Paris2024_EVENTS[sport] and Paris2024_EVENTS[event] to the Field well.
- Format the X-axis type to categorical to avoid automatic date/time conversion.
- Create and display in acard visual a DAX measure that calculates the number of countries involved in the chart:
  NB NOCS in Medals = DISTINCTCOUNT(Paris2024_MEDALS[NOCID])
- Explain why COUNTROWS(Paris2024_NOCS[NOCID]) does not yield the same result.

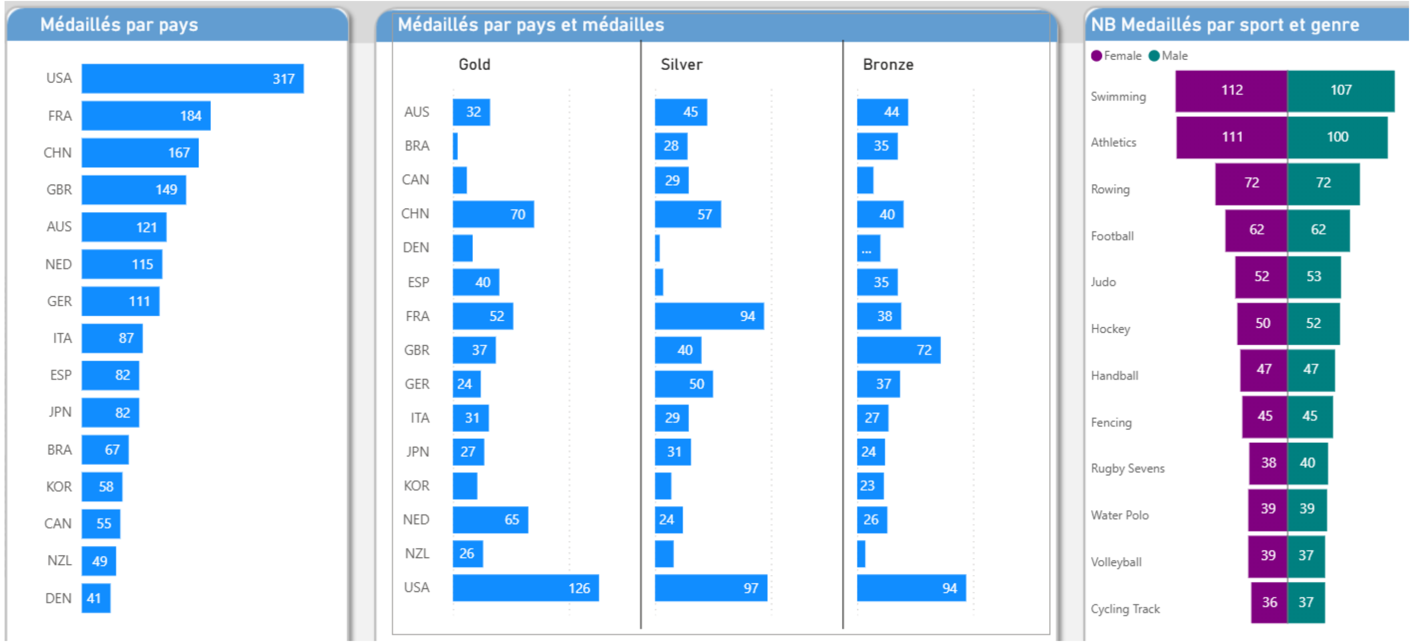
Stacked Histogram



Small multiples and tornado chart

- Insert a grouped bar design with Paris2024_NOCS[code] in "Y-Axis" and [NB Medallists] in "X-Axis".
- Duplicate this visual and add Medal Sort Table[Medal] in "Small Multiples", format in 1 row and 3 columns.
- Add a Tornado visual with Paris2024_EVENTS[sport] in "Group", Paris2024_MEDALLISTS[gender] in "Legend" and [NB Medallists] in "Values" on the right.





CROSS-HIGHLIGHTING / FILTERING

CROSS-HIGHLIGHTING / CROSS-FILTERING

In Power BI, when you click on or select part of a visual (like a bar, line, or point), that interaction can affect the other visuals on the same page in one of two ways:

1. **Cross-filtering** → means that selecting an element in one visual *filters* the data shown in other visuals so that only related records remain visible. Common in visuals like **slicers**, **tables**, and **cards**.

2. **Cross-highlighting** → dims unrelated data in other visuals but keeps all values visible, highlighting only the relevant ones. Works best with **bar**, **column**, or **pie** visuals.

You can define whether a visual **cross-filters** or **cross-highlights** others:
Select a visual → Go to the **Format ribbon** → **Edit interactions**
On each other visual, you'll see icons:

- **Filter icon** → Cross-filter
- **Highlight icon** → Cross-highlight
- **No icon** → No effect

CROSS-HIGHLIGHTING / CROSS-FILTERING

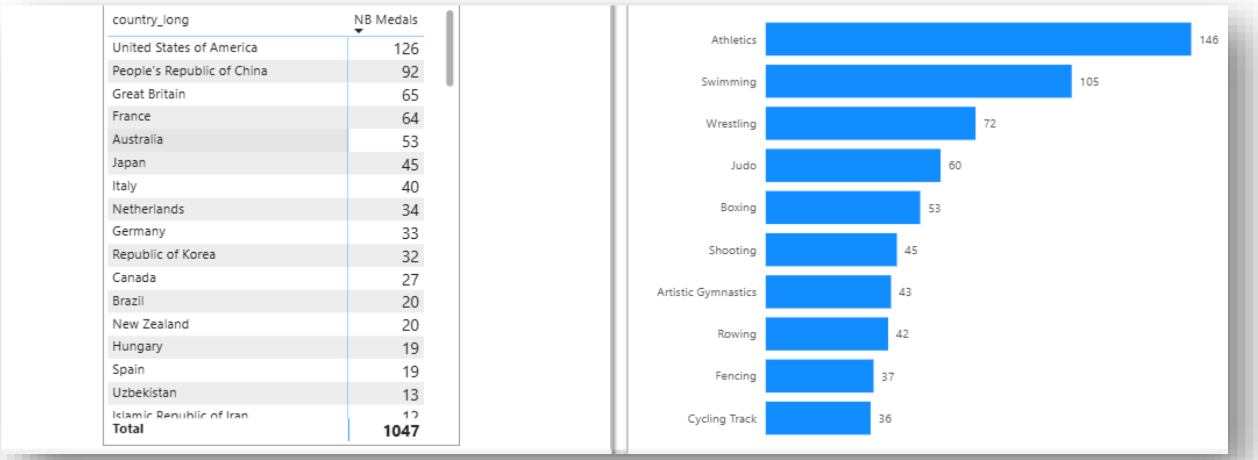
1. Insert a table visual with Paris2024_NOCS[country_long] and [NB medals] in the rows well.

2. Insert a bar chart with dimDisciplines[sport] in the Y-axis well and [NB_medals]in the X-axis well.

3. Duplicate the bar chart.

4. Select the table then edit the interactions (Format > Edit interactions) and click on the filter button on the bar chart .

Click on one of the countries in the table visual to see it cross-higlighting the 1st bar chart and cross-filtering the 2nd one.



Combo chart

Combining lines & columns + average

Combo chart exercise : objective

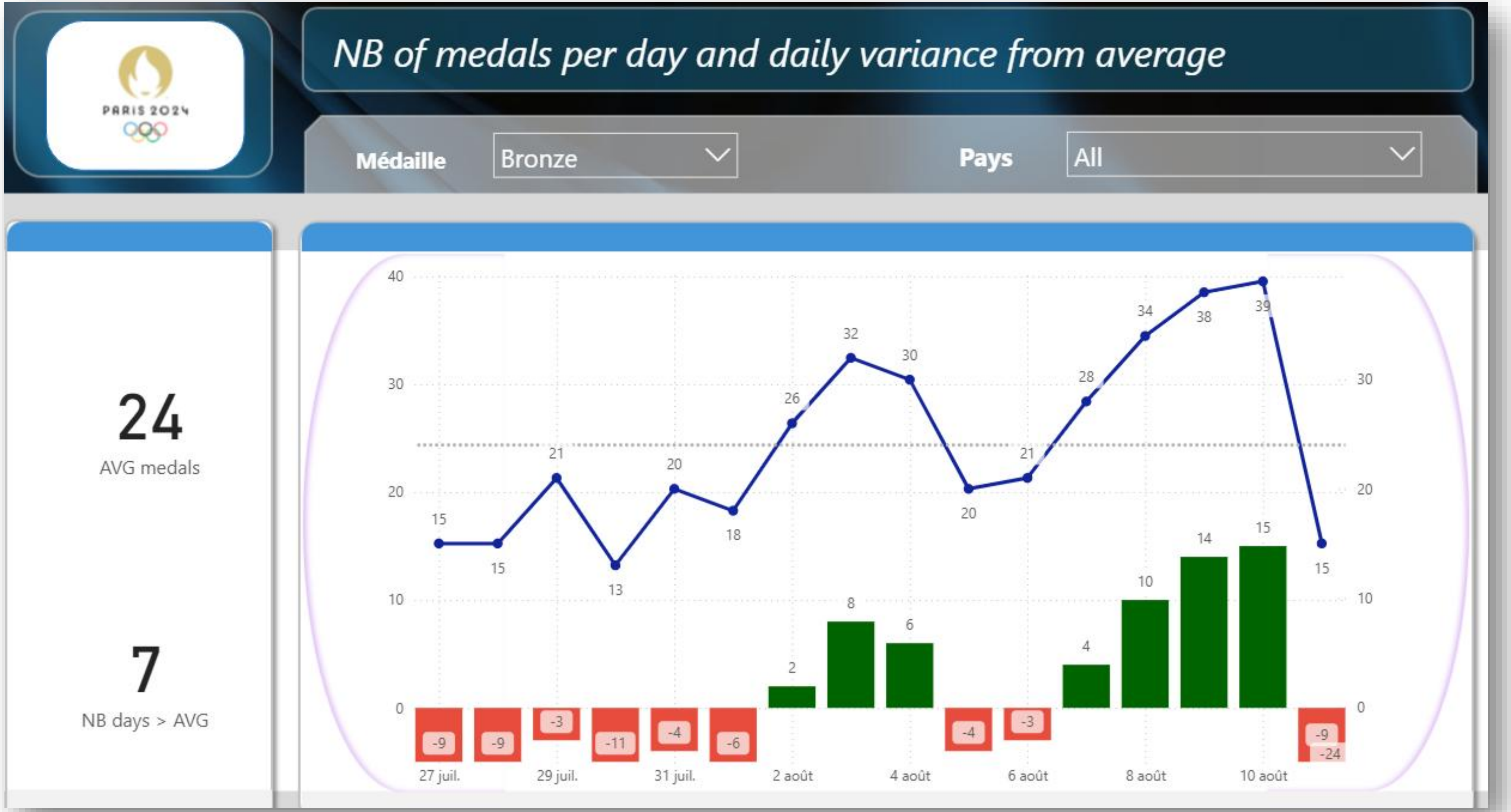
- **GOAL:**
  Create a line chart that displays the number of medals per day. You have to add a line for the global average of medals over all days.

- Add a column chart that will display the daily variation from the global average value.

- There should be a slicer with the list of countries in the header section that filters the results for the selected country.

- You have to add two cards displaying the average number of medals and the number of days where the variation was positive (NB medals > average).

Combo chart



NB of medals per day and daily variance from average
PARIS 2024
Médaille
Bronze
Pays
All
24
AVG medals
7
NB days > AVG
27 juil.
29 juil.
31 juil.
2 août
4 août
6 août
8 août
10 août

Insert a Row/Column Combo Visual

- Insert a combo visual with [NB Medals] in "Line Y-Axis" and [Date] in "X-Axis".
- Create a DAX measure that calculates daily average of medals (keeping external filters active) and add it to "Y Axis of Line":

```
AVG medals =
 VAR __NBmedals = CALCULATE([NB Medals],ALLSELECTED('CALENDAR'[Date]))
 VAR __NBdays = CALCULATE([NB days],REMOVEFILTERS(Paris2024_MEDALS))
 VAR __result = DIVIDE(__NBmedals,__NBdays)
 RETURN
 ROUNDDOWN(__result,0)
```

- Create a 2nd measure that calculates the variation between the number of medals and the average:
  Variation NB medals-AVG = [NB Medals] - [AVG medals]
  Add it to "Y-axis of the column".

Combo chart exercise

- Create a measure that calculates the number of days where the number of medals was greater than the average.

```
NB days > AVG =
 VAR __threshold = [AVG medals]
 VAR __vsz1 = SUMMARIZE(Paris2024_MEDALS,'CALENDAR'[Date],"@NB",[NB Medals])
 VAR __vf1 = FILTER(__vsz1, [@NB] > __threshold)
 VAR __result = COUNTROWS(__vf1)
 RETURN __result
```

- Add 2 card visuals to display [AVG medals] and [NB days > AVG].
- Modify the Y-axis and secondary y-axis min and max values to avoid lines and columns crossing each other.

Combo chart exercise : Solution

- Turn the data labels on and create a DAX measure to apply conditional formatting on the columns colors (red when negative and green when positive):

  Color for variance = IF([Variation NB medals-AVG] >=0, "darkgreen", "#E74C3C")

  Format > Columns > Color > fx then select [Color for variance]

- Add a slicer (drop-down list) on the country and a visual map that displays [AVG Medals].

- Add a slicer (drop-down list) on Medal Sort Table[Medal].

- Insert a scrim background.


Number of Medals by Country (TOP 10)
PARIS 2024
GOLD
SILVER
BRONZE

BOOKMARKS


Power BI
Desktop

Power BI – What is a bookmark ?

In **Power BI**, a **bookmark** is a saved *snapshot* of a report page in its current state — including filters, slicer selections and visuals' visibility,

Typical uses:

•**Interactive navigation** – Create buttons that let users move between filtered views (e.g., "Show Top 10", "Show All Countries").

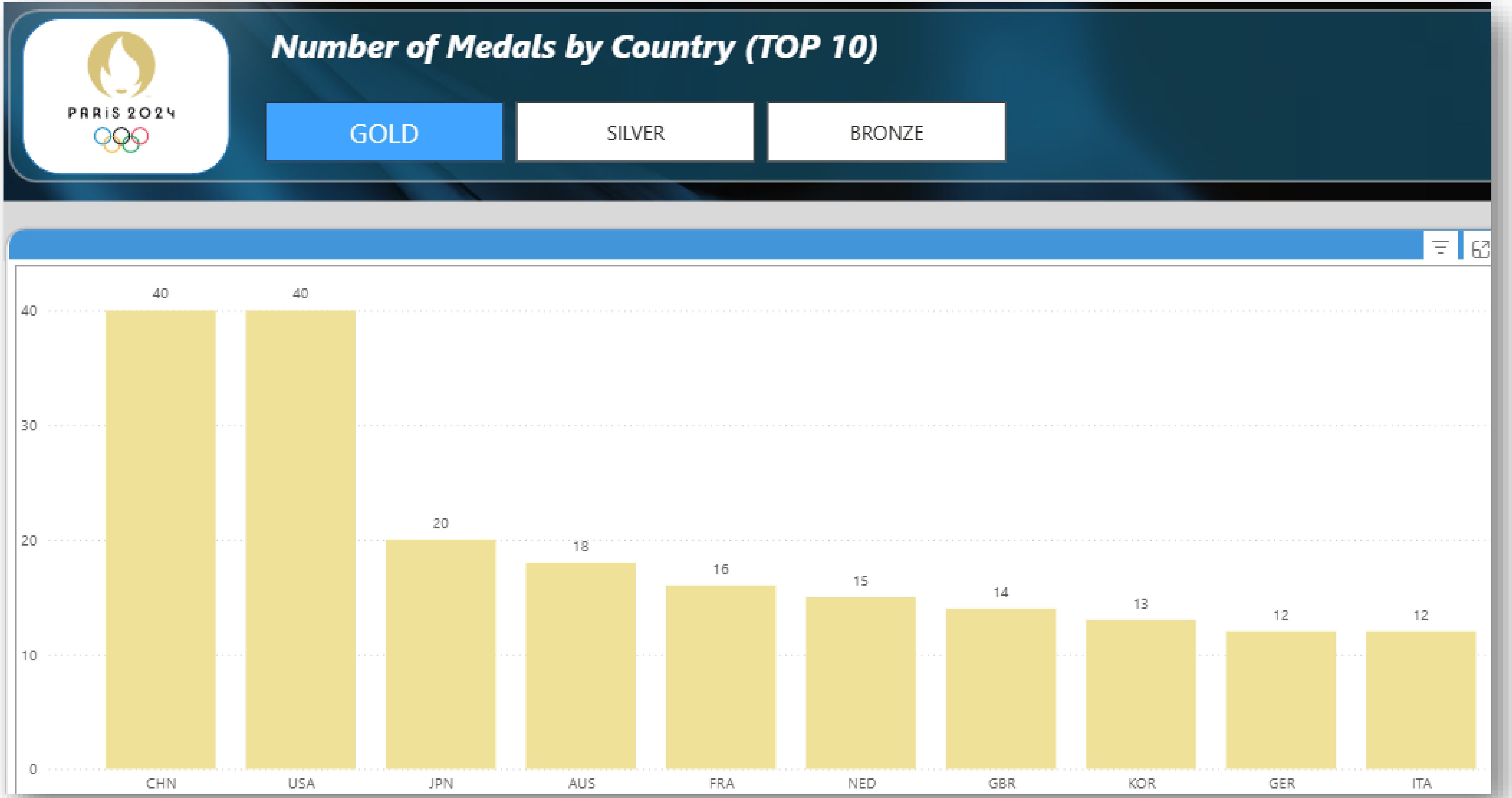
•**Storytelling / Presentations** – walk through insights step-by-step.

•**Custom views** – Allow users to switch between visuals or layouts (e.g., "Map View" vs. "Table View").

Process:

Create several snapshots (bookmarks) of a report page then add a bookmark navigator (Insert > Buttons > bookmark navigator) to display the bookmark of your choice.

Using bookmarks



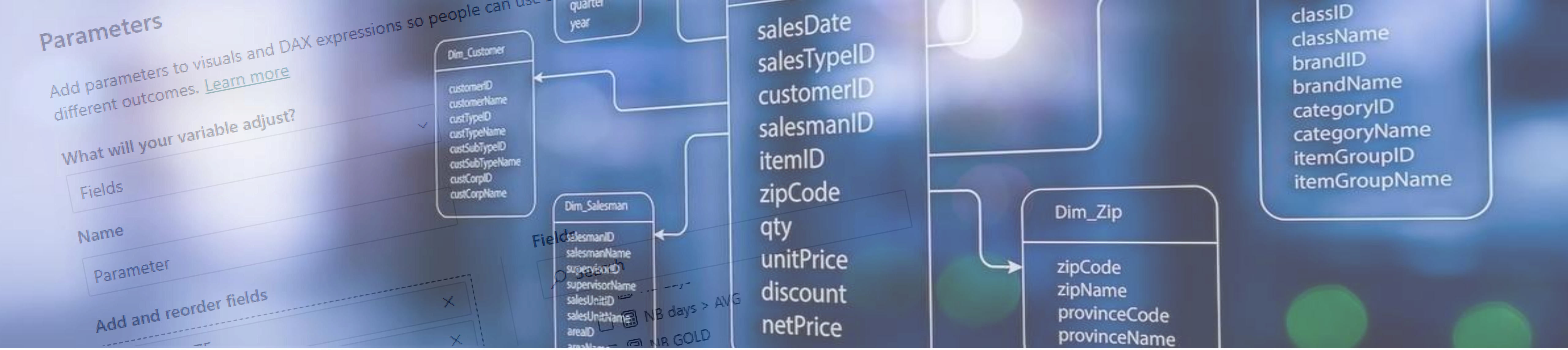
Bookmarks: Exercise

Create 3 DAX measures for calculating the number of Gold, silver and bronze medals.
Insert 3 column charts, one for each measure created with NOCS[code] on the X axis. For each chart , go to Format > Columns > Color and choose a color matching the medal type.

Activate the bookmarks pane (View > bookmarks) then activate the selection pane and only show one visual on the page (hide the others in the selection pane) then save this snapshot as a bookmark. Repeat this process, activating one visual at a time. Create a bookmark group and regroup the 3 bookmarks there.

Insert a bookmarks navigator in the page (Ctrl + click the bookmarks). Make sure that the bookmark navigator only points to the group created earlier (Format > Bookmarks > Bookmark).

Add a scrim in the background and the official logo as an image.



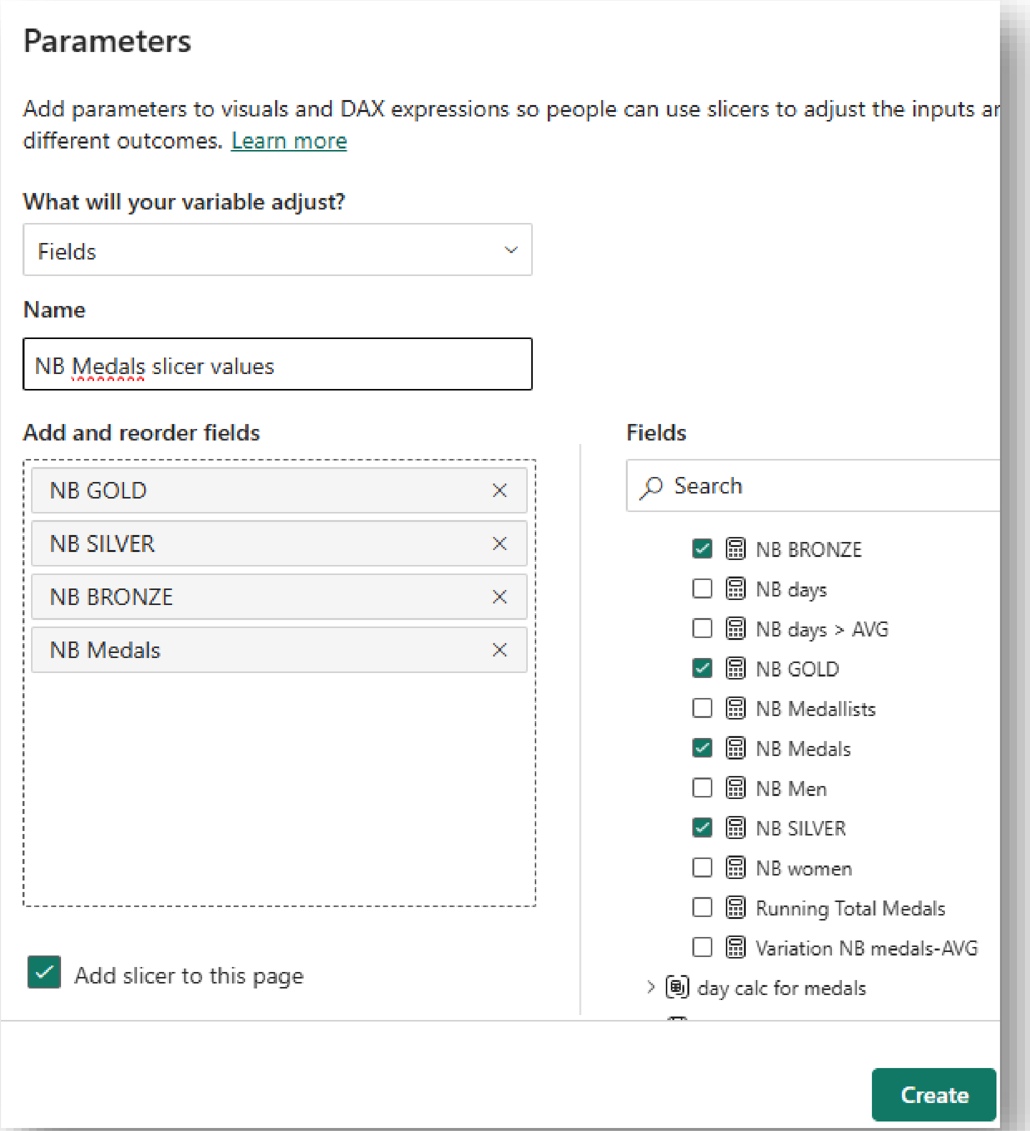
FIELD PARAMETER

Using a field parameter

A **field parameter** in Power BI is a dynamic table that lets users **switch between different fields or measures** in visuals — without needing separate visuals or complex DAX.

Exercise:
On a new page, go to Modeling > New parameter > Fields parameter. Name it ‘NB medals slicer values’ and select the measures as shown on the screenshot.

This creates a new table containing links to the DAX measures and an associated slicer in the active page.



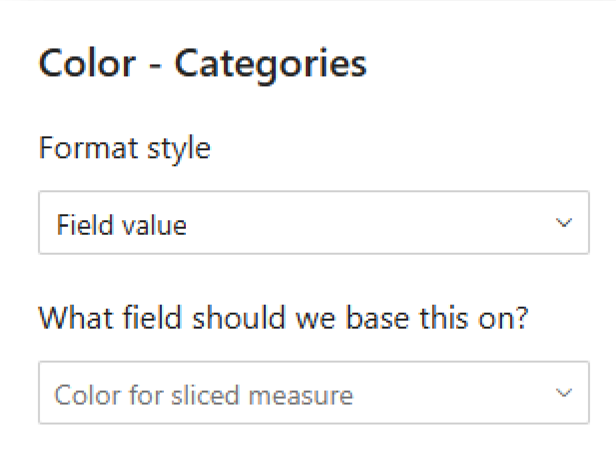
Using a field parameter

Insert a column chart and drag the ‘NB Medals slicer values[NB Medals slicer values]to the Y-axis well and NOCS[code] to the X-axis well.

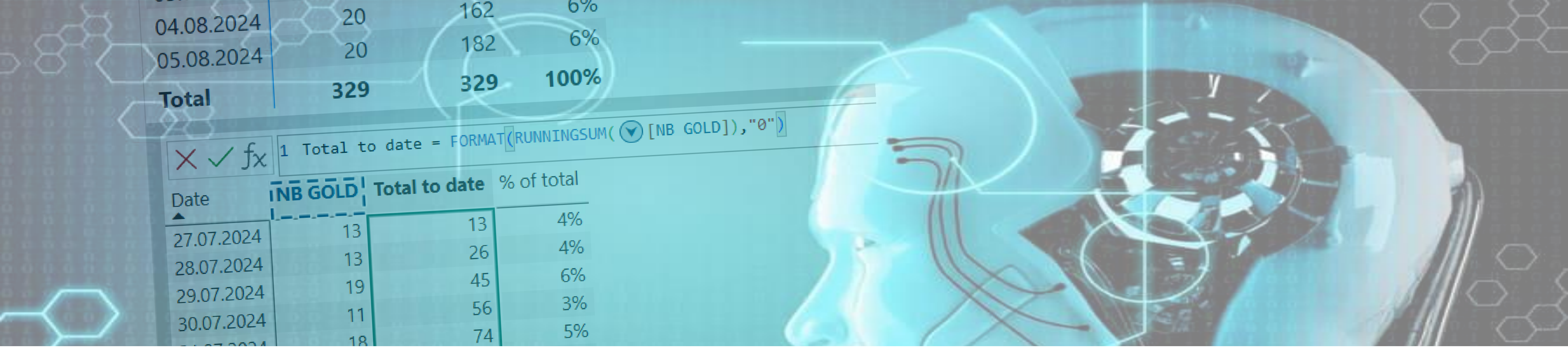
Create a DAX measure that will conditionally format the color of the columns according to the type of medal selected in the slicer as follows:

```
Color for sliced measure =
SWITCH(SELECTEDVALUE('NB medals slicer values'[NB medals slicer values Fields]),
        "'01.Measures'[NB GOLD]","#A38600",
        "'01.Measures'[NB SILVER]","#CCCCCC",
        "'01.Measures'[NB BRONZE]","#F0E199",
        "#0000AA")
```

Assign this measure to the conditional formatting of the column chart:
go to Format > Columns > Color > fx and set the style as ‘Field value’
Then select the measure created above in
‘What field should we base this on ?’



Displaying various measures in one visual


04.08.2024 20 162 6%
05.08.2024 20 182 6%
Total 329 329 100%
1 Total to date = FORMAT(RUNNINGSUM([NB GOLD]),"0")
Date
NB GOLD
Total to date
% of total
27.07.2024 13 13 4%
28.07.2024 13 26 4%
29.07.2024 19 45 6%
30.07.2024 11 56 3%
18 74 5%

Visual DAX


Power BI
Desktop

Adding calculation with visual DAX

- In a new page, add a table visual and drag [NB medals slicer values] in the Values well and [Date] in the rows.
- Insert a slicer based on [NB medals slicer values].
- Click on Home > Insert a new visual calculation
- In the new window that opens, select ' Running sum' by clicking the fx button then fill the formula as follows: Total to date = RUNNINGSUM([NB GOLD]).
- Add the FORMAT() function to display the running total without decimals:
  Total to date = FORMAT(RUNNINGSUM([NB GOLD]),"0")
- Click again on fx and select "percent of grand total' to create another calculation and complete the DAX code as follows:
  % of total = FORMAT(DIVIDE([NB GOLD], COLLAPSEALL([NB GOLD], ROWS)),"0%")

Adding calculation with visual DAX

- Click again on fx and select ‘Versus previous’ to create another calculation and complete the DAX code as follows:

```
Daily change =
    VAR __diff = [NB GOLD] - PREVIOUS([NB GOLD])
    VAR __result = IF(__diff < 0,
                __diff,
                "+"& __diff
                )
    RETURN __result
```

Date	NB GOLD	Total to date	% of total	Daily change
27.07.2024	13	13	4%	+13
28.07.2024	13	26	4%	+0
29.07.2024	19	45	6%	+6
30.07.2024	11	56	3%	-8
31.07.2024	18	74	5%	+7
01.08.2024	16	90	5%	-2
02.08.2024	23	113	7%	+7
03.08.2024	29	142	9%	+6

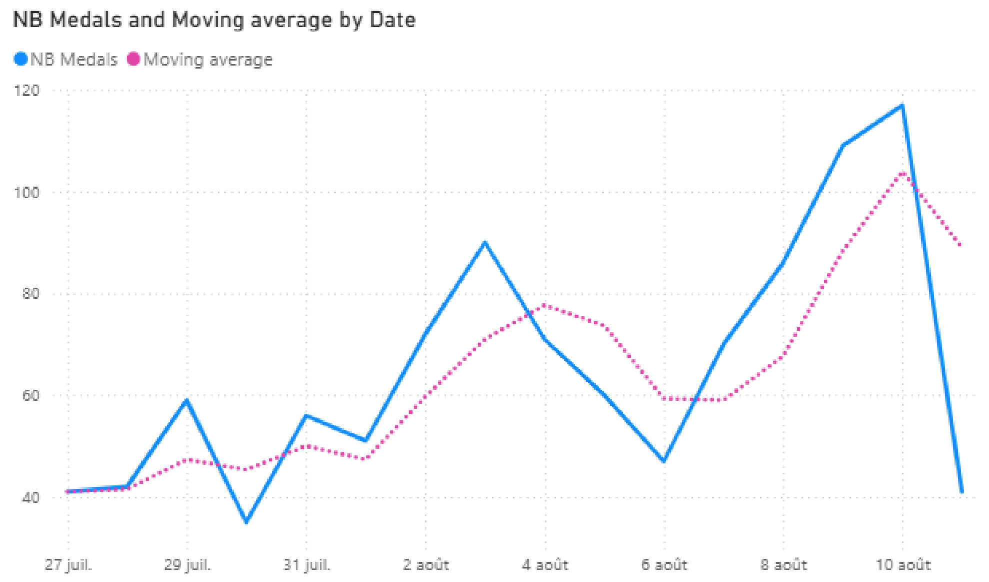
Adding calculation with visual DAX

- Insert another visual calculation and choose moving average then complete DAX code as follows:

  Moving average = MOVINGAVERAGE([NB Medals], 3)

Gold medals statistics using visual calculations

Date	NB GOLD	Total to date	% of total	Daily change
27.07.2024	13	13	4%	+13
28.07.2024	13	26	4%	+0
29.07.2024	19	45	6%	+6
30.07.2024	11	56	3%	-8
31.07.2024	18	74	5%	+7
01.08.2024	16	90	5%	-2
02.08.2024	23	113	7%	+7
03.08.2024	29	142	9%	+6
04.08.2024	20	162	6%	-9
05.08.2024	20	182	6%	+0
06.08.2024	13	195	4%	-7
07.08.2024	21	216	6%	+8
08.08.2024	26	242	8%	+5
09.08.2024	35	277	11%	+9
10.08.2024	39	316	12%	+4
11.08.2024	13	329	4%	-26
Total	**329**	**329**	**100%**	**+329**



Calculation groups

Using calculation groups

- NB BRONZE
- NB SILVER
- NB GOLD (selected)
- TOTAL

Statistics using calculation groups and field parameters

- Current day
- D-1
- D-2
- Var D-1
- AVG 3 days
- % of total
- Running sum

Date	Current day	D-1	D-2	Var D-1	AVG 3 days	% of total	Running sum
27.07.2024	13			100,0%	4	4,0%	13
28.07.2024	13	13		0,0%	8	4,0%	26
29.07.2024	19	13	13	31,6%	15	5,8%	45
30.07.2024	11	19	13	-72,7%	14	3,3%	56
31.07.2024	18	11	19	38,9%	16	5,5%	74
01.08.2024	16	18	11	-12,5%	15	4,9%	90
02.08.2024	23	16	18	30,4%	19	7,0%	113
03.08.2024	29	23	16	20,7%	22	8,8%	142
04.08.2024	20	29	23	-45,0%	24	6,1%	162
05.08.2024	20	20	29	0,0%	23	6,1%	182
06.08.2024	13	20	20	-53,8%	17	4,0%	195
07.08.2024	21	13	20	38,1%	18	6,4%	216
08.08.2024	26	21	13	19,2%	20	7,9%	242
09.08.2024	35	26	21	25,7%	27	10,6%	277
10.08.2024	39	35	26	10,3%	33	11,9%	316
11.08.2024	13	39	35	-200,0%	29	4,0%	329
Total	**329**	**316**	**277**	**4,0%**	**307**	**100,0%**	**329**

Calculation group: Set up

- Go to Model view and in the Data pane, select the 'Model' tab and right click on 'Calculation groups' > New calculation group.
- Rename the default calculation item to 'Current day'.
- Right click on 'Calculation item' > New calculation item and insert the following DAX code:

```
D-1 =
 VAR __prevday = CALCULATE(SELECTEDMEASURE(), DATEADD('CALENDAR'[Date],-1,DAY))
 VAR __result = IF(NOT ISBLANK(__prevday),__prevday,"")
 RETURN __result
```

- Repeat with:

```
D-2 =
VAR __minus2days = CALCULATE(SELECTEDMEASURE(), DATEADD('CALENDAR'[Date],-2,DAY))
VAR __result = IF(NOT ISBLANK(__minus2days),__minus2days,"")
RETURN __result
```

- Add another calculation item:

```
Var D-1 =
    VAR __previous = CALCULATE(SELECTEDMEASURE(),DATEADD('CALENDAR'[Date],-1,DAY))
    VAR __diff = SELECTEDMEASURE() - __previous
    VAR __result = DIVIDE(__diff, SELECTEDMEASURE())
    RETURN
    IF( ISBLANK(SELECTEDMEASURE()) ||  ISBLANK(__previous)),
       BLANK(),
       FORMAT( __result,"0.0%")
    )
```

Calculation group: Set up

- Add the 3 days average calculation item :

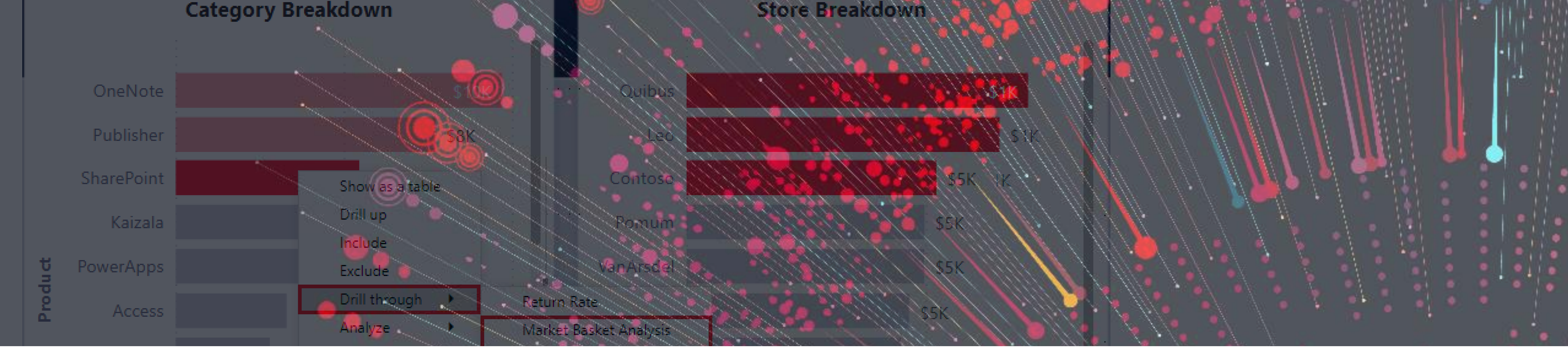
```
AVG 3 days =
    VAR __currday = SELECTEDMEASURE()
    VAR __prevday = CALCULATE(SELECTEDMEASURE(),DATEADD('CALENDAR'[Date],-1,DAY))
    VAR __minus2day = CALCULATE(SELECTEDMEASURE(),DATEADD('CALENDAR'[Date],-2,DAY))
    VAR __result = ROUNDDOWN(DIVIDE(__currday+__prevday+__minus2day,3),0)
    return __result
```

- Create the ‘% of total’ calculation item:

```
% of total =
    VAR __pct =
        DIVIDE(
            SELECTEDMEASURE(),
            CALCULATE( SELECTEDMEASURE(), ALL(CALENDAR[Date])
        ))
    VAR __result = FORMAT(__pct,"0.0%")
    RETURN __result
```

Calculation group: Set up

- Rename the calculation group ‘day calc for medals’
- Go back to report view, create a new page and insert a table visual with Calendar[Date] in the Rows and the calculation group column in the Columns. Drag the field parameter created earlier [NB medals slicer value] in the values.
- Insert a button slicer with the calculation group column again. The slicer now shows all calculation items available and you can select one calculation or more to be displayed. The medals slicer allows you to view the different calculations over the selected measure.

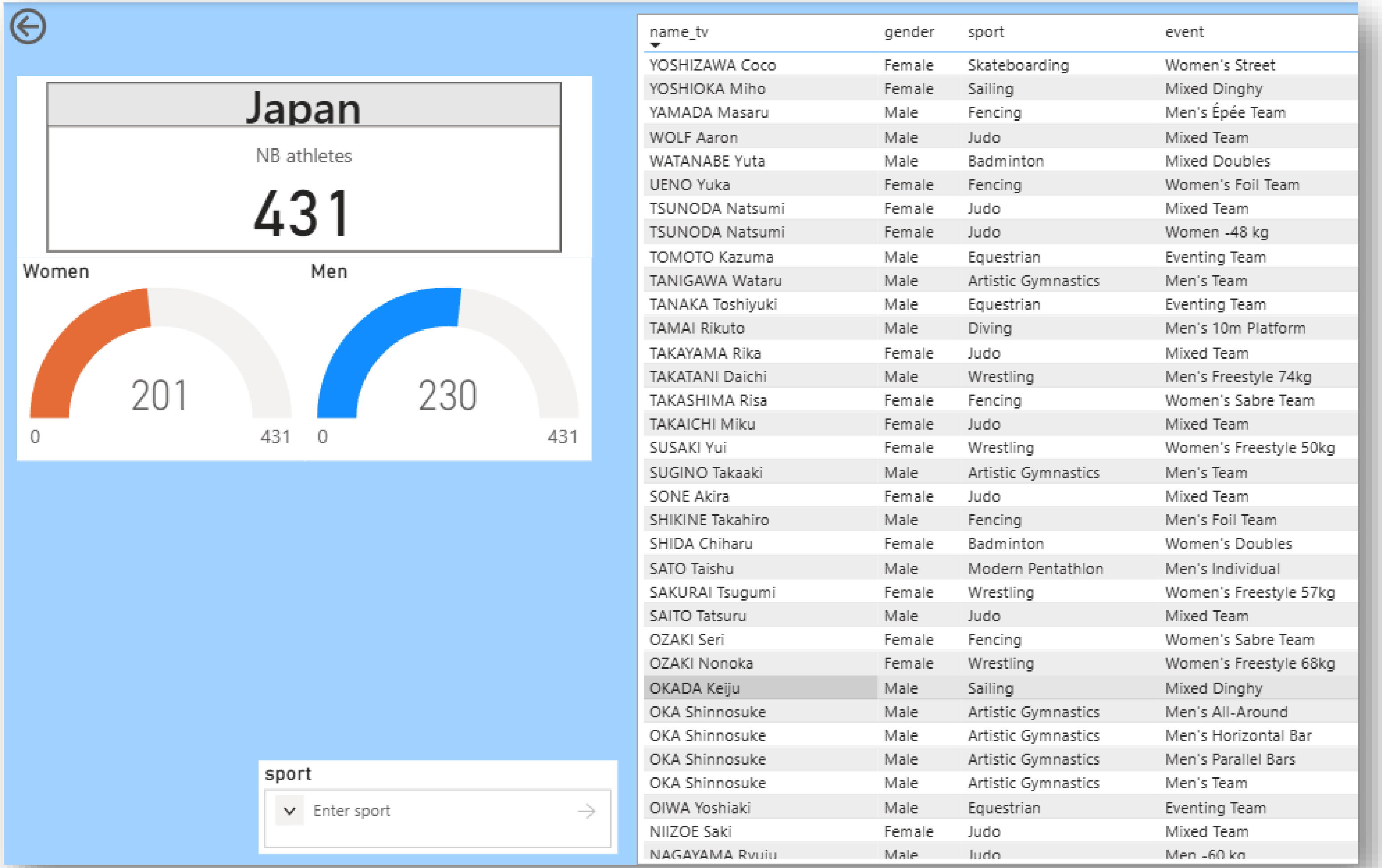


DRILLTHROUGH PAGE

Drillthrough page

- Create a new page and in the page format section, select ‘drillthrough’ under Page Type then drag NOCS[country_long], NOCS[code] and NOCS[nocID] in the ‘Drill though from’ well.
- Add a card visual with [NB Athletes] in the Value well and [country_long] in the Categories.
- Add 2 gauges visuals with [NB Men] & [NB Women] and [NB Athletes] as the maximum value .
- Add a table visual and drag the following fields in the ‘Columns’ well : Paris2024_ATHLETES[name_tv], Paris2024_ATHLETES[gender], Paris2024_EVENTS[sport], Paris2024_EVENTS[event].
- Add an input (text) slicer to the page and drag Paris2024_EVENTS[sport] in the Field well. You can now search for specific values in the table.
- Wherever the country name or code is used in any visual of the report, the drillthrough option will now be enabled and will point to the current page.
- Change the background color of the page to make it stand out.

Drillthrough page

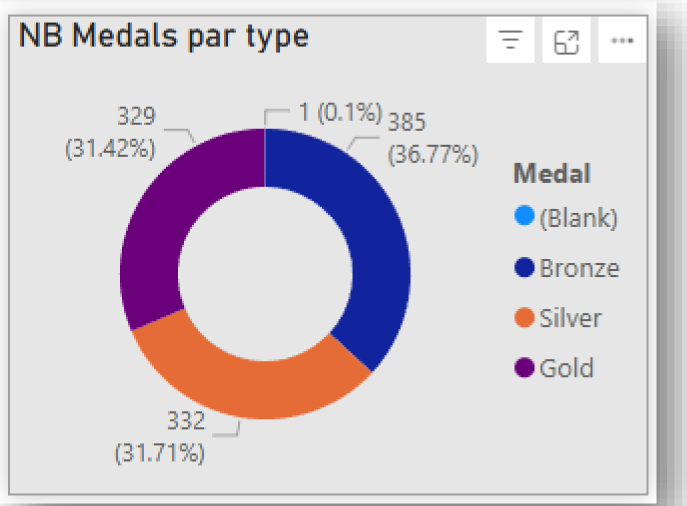


name_tv	gender	sport	event
YOSHIZAWA Coco	Female	Skateboarding	Women's Street
YOSHIOKA Miho	Female	Sailing	Mixed Dinghy
YAMADA Masaru	Male	Fencing	Men's Épée Team
WOLF Aaron	Male	Judo	Mixed Team
WATANABE Yuta	Male	Badminton	Mixed Doubles
UENO Yuka	Female	Fencing	Women's Foil Team
TSUNODA Natsumi	Female	Judo	Mixed Team
TSUNODA Natsumi	Female	Judo	Women -48 kg
TOMOTO Kazuma	Male	Equestrian	Eventing Team
TANIGAWA Wataru	Male	Artistic Gymnastics	Men's Team
TANAKA Toshiyuki	Male	Equestrian	Eventing Team
TAMAI Rikuto	Male	Diving	Men's 10m Platform
TAKAYAMA Rika	Female	Judo	Mixed Team
TAKATANI Daichi	Male	Wrestling	Men's Freestyle 74kg
TAKASHIMA Risa	Female	Fencing	Women's Sabre Team
TAKAICHI Miku	Female	Judo	Mixed Team
SUSAKI Yui	Female	Wrestling	Women's Freestyle 50kg
SUGINO Takaaki	Male	Artistic Gymnastics	Men's Team
SONE Akira	Female	Judo	Mixed Team
SHIKINE Takahiro	Male	Fencing	Men's Foil Team
SHIDA Chiharu	Female	Badminton	Women's Doubles
SATO Taishu	Male	Modern Pentathlon	Men's Individual
SAKURAI Tsugumi	Female	Wrestling	Women's Freestyle 57kg
SAITO Tatsuru	Male	Judo	Mixed Team
OZAKI Seri	Female	Fencing	Women's Sabre Team
OZAKI Nonoka	Female	Wrestling	Women's Freestyle 68kg
OKADA Keiju	Male	Sailing	Mixed Dinghy
OKA Shinnosuke	Male	Artistic Gymnastics	Men's All-Around
OKA Shinnosuke	Male	Artistic Gymnastics	Men's Horizontal Bar
OKA Shinnosuke	Male	Artistic Gymnastics	Men's Parallel Bars
OKA Shinnosuke	Male	Artistic Gymnastics	Men's Team
OIWA Yoshiaki	Male	Equestrian	Eventing Team
NIIZOE Saki	Female	Judo	Mixed Team
NAGAYAMA Ryuju	Male	Judo	Men -60 kg

TOOLTIP PAGE

Tooltip page

- Create a new page and in the page format section, select 'drillthrough' under Page information > Page Type then drag NOCS[country_long], NOCS[code] and NOCS[nocID] in the 'Drill though from' well.
- Add a donut visual with [NB Medals] in the Value well and Medal Sort Table[Medal] in the Legend.
- Change the background color of the page to create contrast.
- This tooltip page will not display every time you hover the mouse over a visual that contains one of the 3 fields from the NOCS table.



Introduction to DAX QUERY VIEW

DAX QUERY VIEW

What is DAX Query View?

- A dedicated environment in Power BI Desktop to write, run, and analyze DAX queries
- Based on DAX Query Language (EVALUATE, VAR, RETURN)
- Allows querying the semantic model without creating visuals

What It Is NOT:

- ❌ It does not replace visuals
- ❌ It does not store results in the model
- ❌ End users never see it — developer tool only

Main Goal: Explore, test, and debug DAX logic directly against the data model

Create your first query

- Insert the following code in a new query and click 'Run':

```
EVALUATE
SUMMARIZE(Paris2024_MEDALS,'CALENDAR'[Date],
"NB",[NB Medals])
```

- Take the code to the next level by adding this code under the previous one:

```
EVALUATE
ADDCOLUMNS(
    SUMMARIZE(Paris2024_MEDALS, 'CALENDAR'[Date]),
    "@NB",[NB Medals],
    "@NB GOLD",[NB GOLD],
    "@pctgold",FORMAT(DIVIDE([NB GOLD],[NB Medals]),"0.0%"),
    "@AVG",[AVG medals]
)
```

Using DAX query view

- View the table used in the combo chart measure:

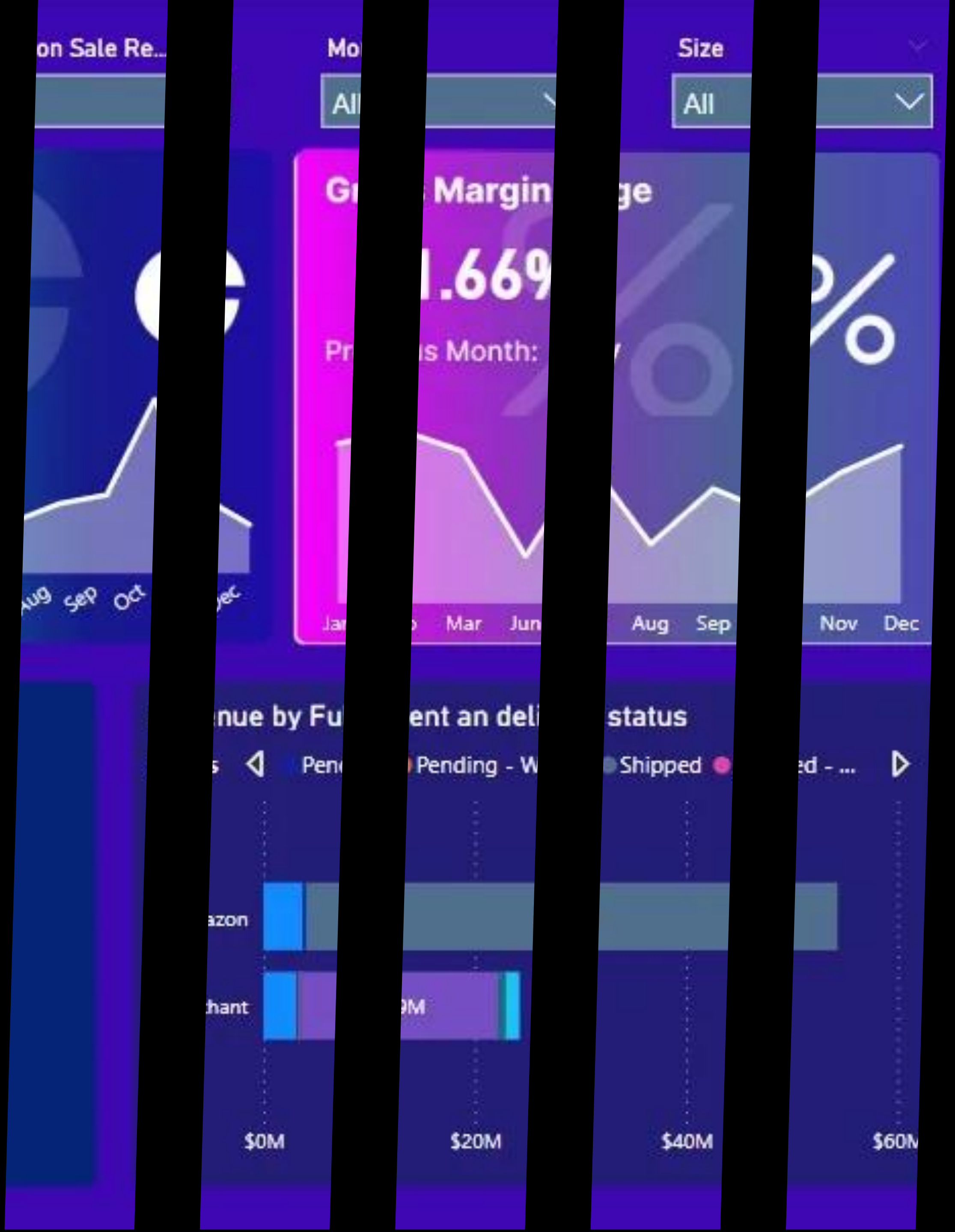
```
EVALUATE

VAR __vt1 =
    ADDCOLUMNS(
SUMMARIZE(Paris2024_MEDALS,'CALENDAR'[Date]),
"@NB",[NB Medals]
)

VAR __avg = [AVG medals]

VAR __vfilt1 = FILTER(__vt1,[@NB] > __avg)

RETURN __vfilt1
```


Size
All
All
Aug
Sep
Mar
Nov
Dec
Shipped
$0M
$20M
$40M


That's all Folks!